Editor's Suggestion

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Rapid Communication

Coaxial cable properties: determining the speed of signal and electric permittivity

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(Received January 25, 2024, Published online February 6, 2024)

The speed of signal in a coaxial cable can be determined by measuring the time delay between varying lengths of coaxial cables that are wired from a signal generator to an oscilloscope. The oscilloscope will show the time delay between the signals passing through each cable. Through many trials of varying cable length differences, a plot of length difference versus time delay can be generated. The slope of the regression line, adjusted to be in ft/ns, represents the speed of signal in the cable. This speed was determined to be 0.69 ± 0.04 ft/ns. The speed is used to calculate the cable's relative electric permittivity, which was determined to be 2.2 ± 0.2.

I. INTRODUCTION

The speed of light is one of the most important physics constants as it represents the upper speed limit that anything can travel in the universe. The speed of light also represents the speed of signal if the signal is traveling in transverse electromagnetic mode (TEM). If this is the case, the accepted value for the speed of signal is 3×10^8 m/s or about 1 ft/ns [1]. However, this only represents the speed of signal in a vacuum. The speed of signal depends on the medium it is traveling through because particles can get in the way and slow the signal down. Therefore, if we were to send a signal through a coaxial cable, the speed of propagation through the medium would be less than its speed in a vacuum.

A coaxial cable is an electrical cable that transmits radio frequencies [2]. These cables consist of an inner conductor that is a standard wire. Around the inner conductor is a dielectric insulator. This insulator has a low electric permittivity, or low polarization in response to an electric field. Around the insulator is the outer conductor, or the shield, which prevents external electrical interference. Lastly, the outer layer of the coaxial cable is a plastic jacket that provides insulation. The inner conductor is responsible for carrying the signal, but the presence of the insulator slows down the signal. For coaxial cables, the signals are transmitted in TEM mode so the equivalency of the speed of signal and light is valid.

Coaxial cables are used as transmission lines for a lot of different applications. These include telephone lines, cable television, and radio transmitters. These cables are used because they can carry high frequencies and have low data loss. Also, the cable design protects the signals from exterior electromagnetic interference. This allows them to have a good signal quality no matter what devices they are placed next to. In addition, knowing the speed of signal within coaxial cables is important for tasks that require proper timing and synchronization.

II. SPEED OF SIGNAL

In a coaxial cable, the speed of signal can be represented as the speed of signal in a dielectric which is

$$\text{signal speed} = \frac{c}{\sqrt{\epsilon_r \mu_r}} \tag{1}$$

where c is the speed of light, ϵ_r is the relative electric permittivity of the insulator, and μ_r is the relative magnetic permeability of the insulator [3]. The values ϵ_r and μ_r are relative to the respective values in a vacuum. Because the insulator does not conduct electrical current, it does not respond to magnetic fields. Therefore, we can take μ_r to be equal to one. Knowing the speed of light c, we can solve for ϵ_r after we determine the speed of signal in the cable.

To conduct the experiment, an oscilloscope and signal generator are used. The cables used were in lengths of 5 ft and 40 ft. The cables could be connected in different pairings to generate different length differences. The signal generator was set to a frequency of 2200 kHz. This frequency was chosen because it was just high enough to show distinguished wave shifts, even with a small cable length difference. For all pairings of cables, each was attached to the signal generator via a T connector and attached to the oscilloscope via terminators. The terminators were required because they allowed the signal to end without bouncing back through the cable and causing interference. Once everything was correctly connected, the time delay was measured with the oscilloscope. This was done by using cursors to measure the time delay between equivalent peaks on the waveforms.

The various length differences and their respective time delays are plotted in fig. 1. A least squares regression line is fitted to the data and adjusted to go through the origin because if there is no length difference between the cables, there should be no time delay. In the case of our experiment, the time delay represents the difference in propagation time it takes each signal to reach the oscilloscope. This difference in time can be related to the length

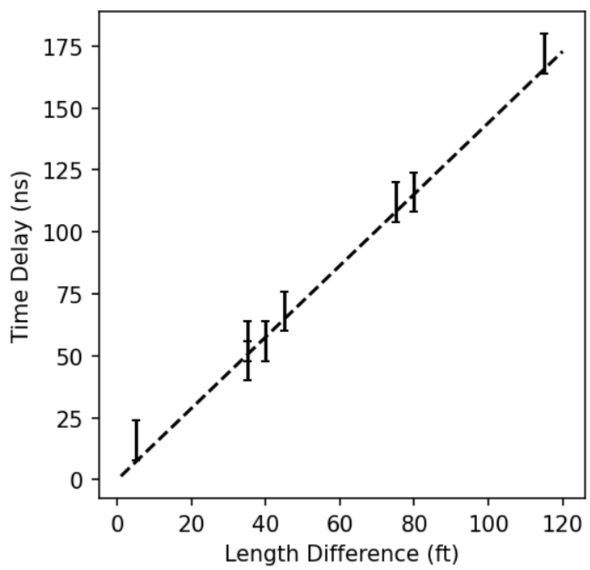

FIG. 1. The length differences and respective time delays are plotted against each other. The least squares linear regression line is adjusted to go through the origin. The inverse of the slope of the regression line represents the speed of signal in the coaxial cables.

difference of the cables, therefore creating a distance and time or a speed relationship. While the units of the slope are ns/ft, the inverse can be taken to get units for speed. Therefore, the inverse of the slope represents the speed of signal in the cable and was found to be

$$\text{signal speed} = 0.69 \pm 0.04 \text{ ft/ns.} \tag{2}$$

This value is about 2/3 the accepted speed of signal in a vacuum of 1 ft/ns. Our result matches our prediction that the speed of signal in a coaxial cable would be less than the speed of signal in a vacuum due to the cable's material density. The particles in the cable interfere with the signal and cause a slower speed. The uncertainty in our result is relatively small, so we can be confident that the speed of the signal is within the 2/3 range of the speed in a vacuum.

Further analysis of the speed of signal in the coaxial cable allows us to determine the relative electric permittivity ϵ_r of the cable mentioned in eq. (1). Given the values for c and μ_r, we can rearrange eq. (1) to solve for ϵ_r and compute that

$$\epsilon_r = 2.2 \pm 0.2. \tag{3}$$

This value is greater than one, but relatively small. The small value is due to the insulator rejecting the formation of an electric field, which is a good property for an insulator to have.

III. CONCLUSION

Through this experiment, we were able to determine the speed of the signal in a coaxial cable through the use of a signal generator and an oscilloscope. By varying the length differences between the cables, we were able to measure the respective time delays. These two variables, length difference and time delay, show a speed relationship and can be plotted against each other. The inverse of the slope of the plot's regression line represents the speed of signal in the cable and was calculated to be 0.69 ± 0.04 ft/ns. Knowing the speed of signal, the relative electric permittivity ϵ_r was calculated to be 2.2 ± 0.2.

Our results were accurate and precise, and they agreed with our predictions. The findings are significant due to their real-world applications. Knowing the speed of sound in a cable is crucial for synchronization and timing in electric circuits. It also ensures accurate measurements when coaxial cables are used in scientific experiments. Knowing the electric permittivity helps in building better electrical devices. Future research could be done to determine signal attenuation based on the length of the cables. This experiment could also be extended to determine the speed of signal and electric permittivity in cables made of different dielectric materials.

Appendix A: Uncertainty calculations

The only uncertainty we considered was the uncertainty in time delay. We did not consider uncertainty in length difference because the lengths of the coaxial cables were given on the cables themselves, so it was assumed they were accurate. The uncertainty in time delay was based on the uncertainty of the oscilloscope. For our setup, each oscilloscope cursor movement resulted in a change of 4 ns. Because we used two cursors to measure time delay, we determined the total uncertainty in time delay σ_t to be 8 ns. To determine the uncertainty in the slope δ_m, we used the equation

$$\delta_m^2 = \frac{1}{\varsigma_x^2 W} \tag{A1}$$

where

$$\varsigma_x^2 := \langle (x - \bar{x})^2 \rangle \tag{A2}$$

$$W := \sum w_i. \tag{A3}$$

and ς_x describes the variance in the x variable, or in our case the length difference. W describes the sum of all weights. The individual weights w_i are each given by $1/\sigma_t^2$ [4].

However, because we take the inverse of the slope to be the speed of the cable, the uncertainty in the slope propagates through the inverse calculation. Therefore, if the inverse of slope m is taken, we find the uncertainty in speed δ_s to be

$$\delta_s = \frac{\delta_m}{m^2} = 0.04 \text{ ft/ns}. \tag{A4}$$

Once the speed s and corresponding uncertainty δ_s are known, eq. (1) can be solved for ϵ_r. Then, by taking the derivative with respect to the speed and multiplying by the uncertainty in speed, the uncertainty in electric permittivity δ_{ϵ_r} is determined to be

$$\delta_{\epsilon_r} = \frac{d(c^2/s^2\mu_r)}{ds}\delta_s = 0.2. \tag{A5}$$

ACKNOWLEDGEMENTS

I would like to thank Sophia Figura for being my lab partner for this experiment. I would also like to thank Justin Wheeler for his help in using the oscilloscope.

[1] NIST, Meet the constants (2019).
[2] T. Cooke, The Global Cable Industry: Materials, Markets, Products , 331 (2021).
[3] U. of Cambridge, The dielectric constant and the refractive index.
[4] J. Anderson, Error propogation and least squares regression (2024).